International Institute for Applied Systems Analysis
IIASA

Crowdsourcing ground biomass data with the IIASA Tree-Quest app and possible scale-up for the Biomass Cal/Val activities

Measuring the World's Forests: Cal/Val Projects for Biomass, Structure, and Disturbance

Milutin Milenković (milenkovic@iiasa.ac.at)

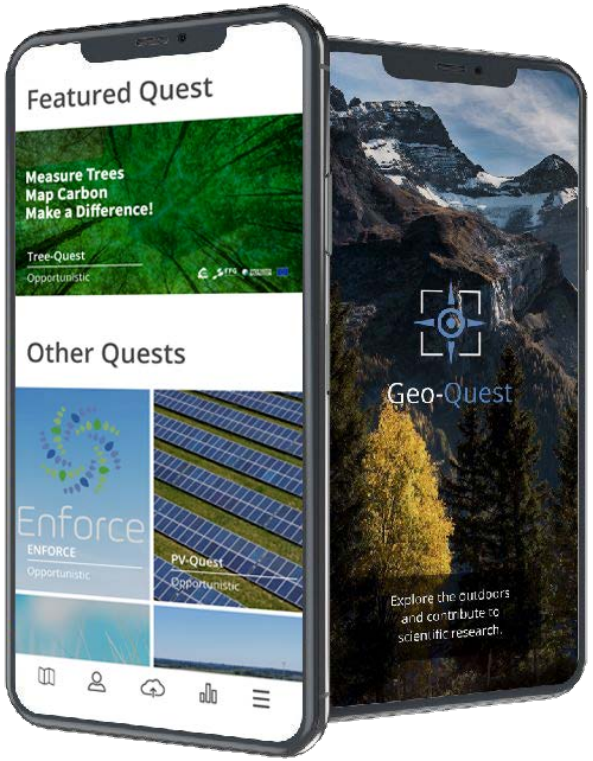
The Novel Data Ecosystems for Sustainability group, IIASA

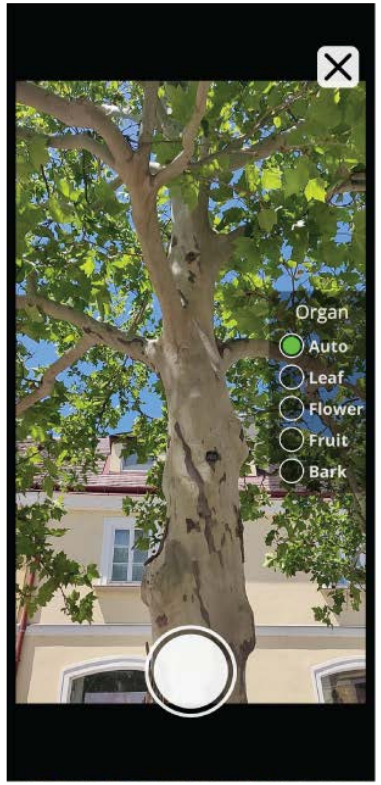
Oct. 22. 2026

FFG Forschung wirkt

Acknowledgment:
CoNSENSAR - Copernicus and National Service Evolution with Next generation SAR missions

Tree-Quest

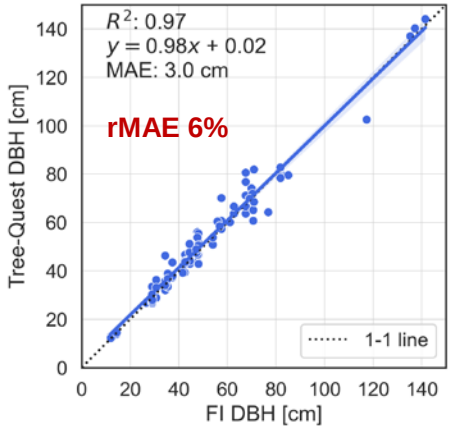


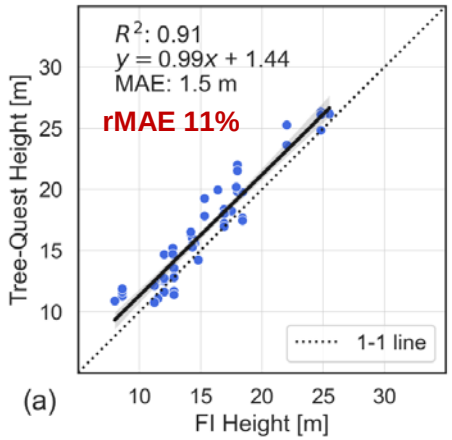






Milenković et al. (2026)
in Ecological Informatics





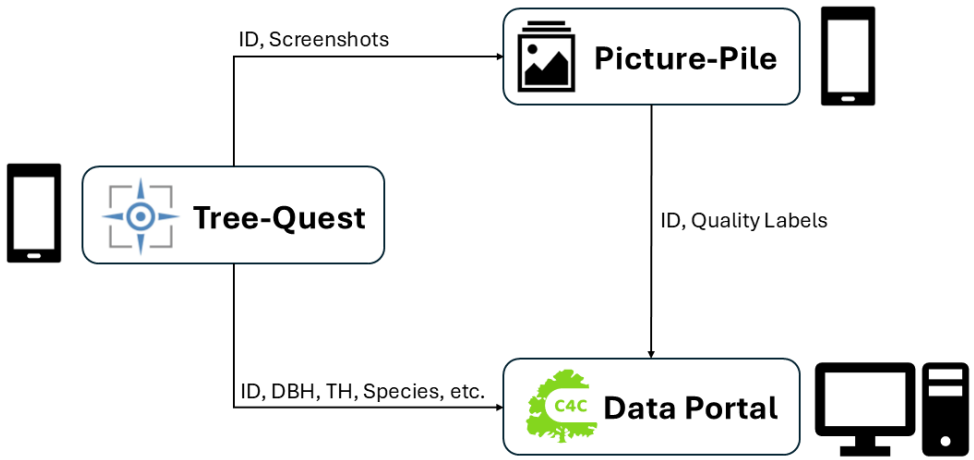
Geo-Quest is a mobile app for **collecting high-quality ground reference data** to improve AI models and remote-sensing applications

https://geoquest.iiasa.ac.at/

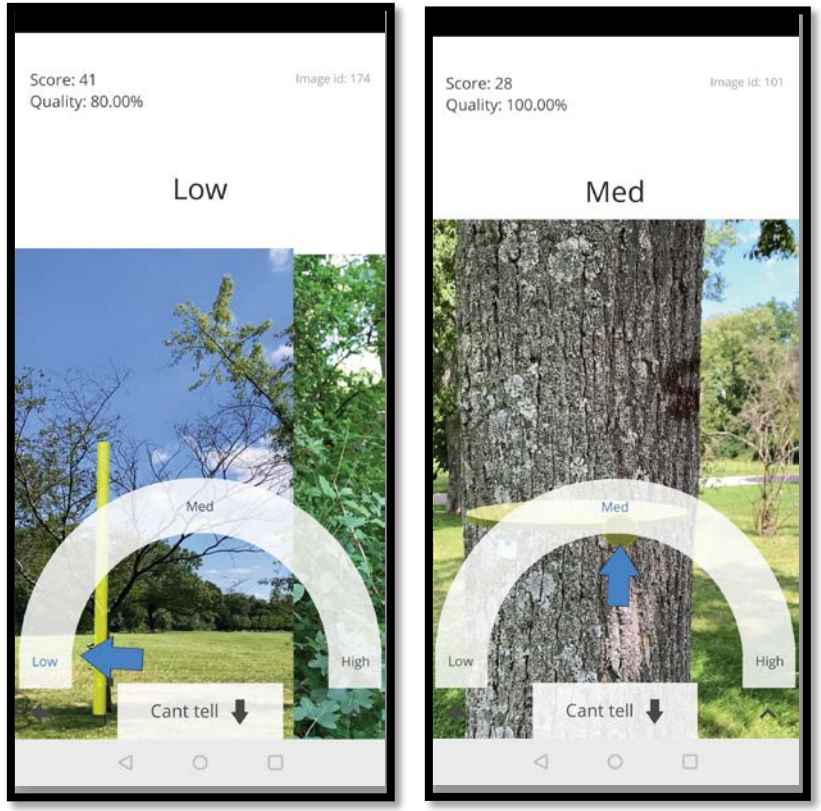
Geo-Quest

Crowdsourcing Workflow

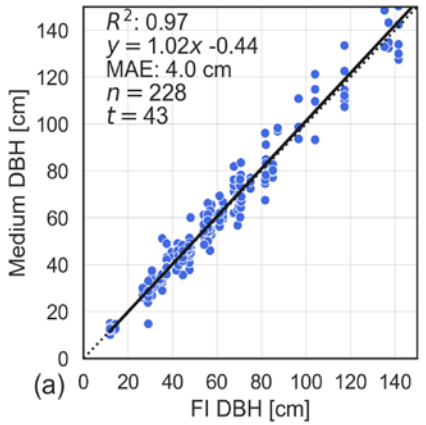




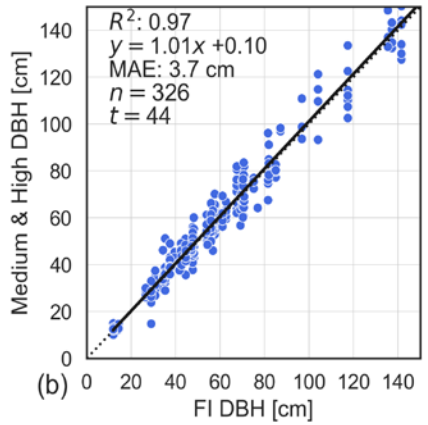
Quality Labelling



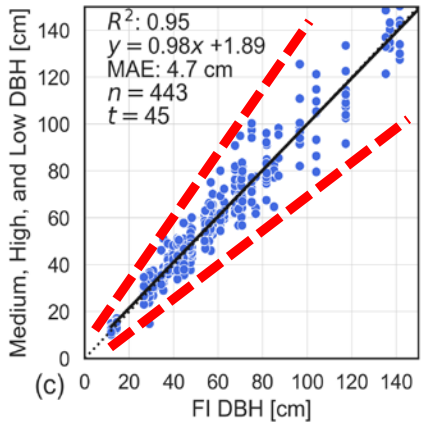
High Quality



High and Medium



High, Medium, and Low



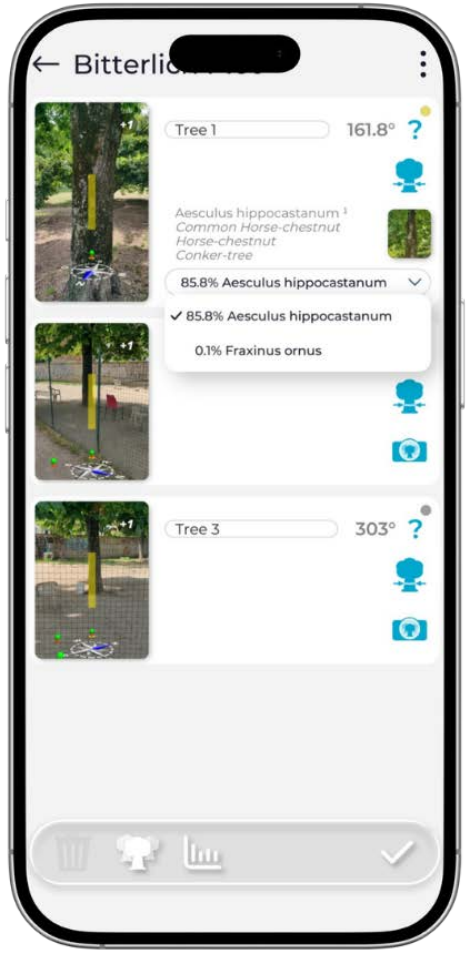
Plot-Quest: Crowdsourcing Plot-Level Information

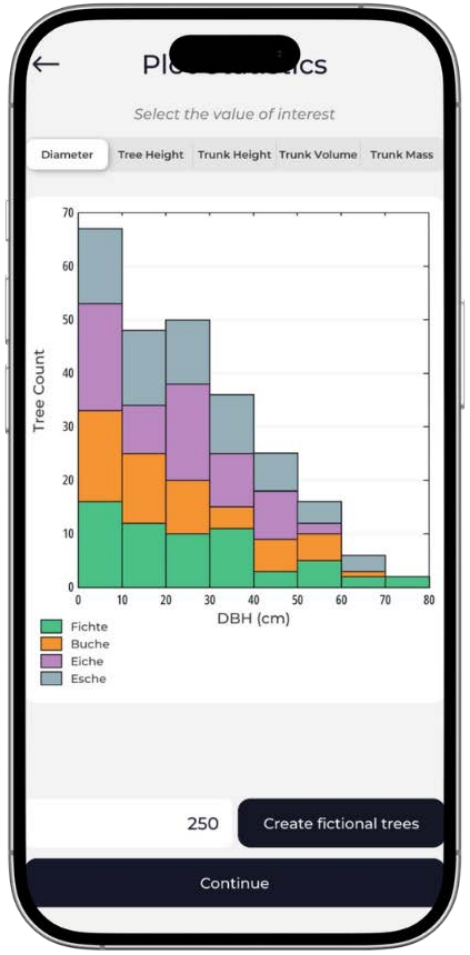
Angle count sampling (ACS)

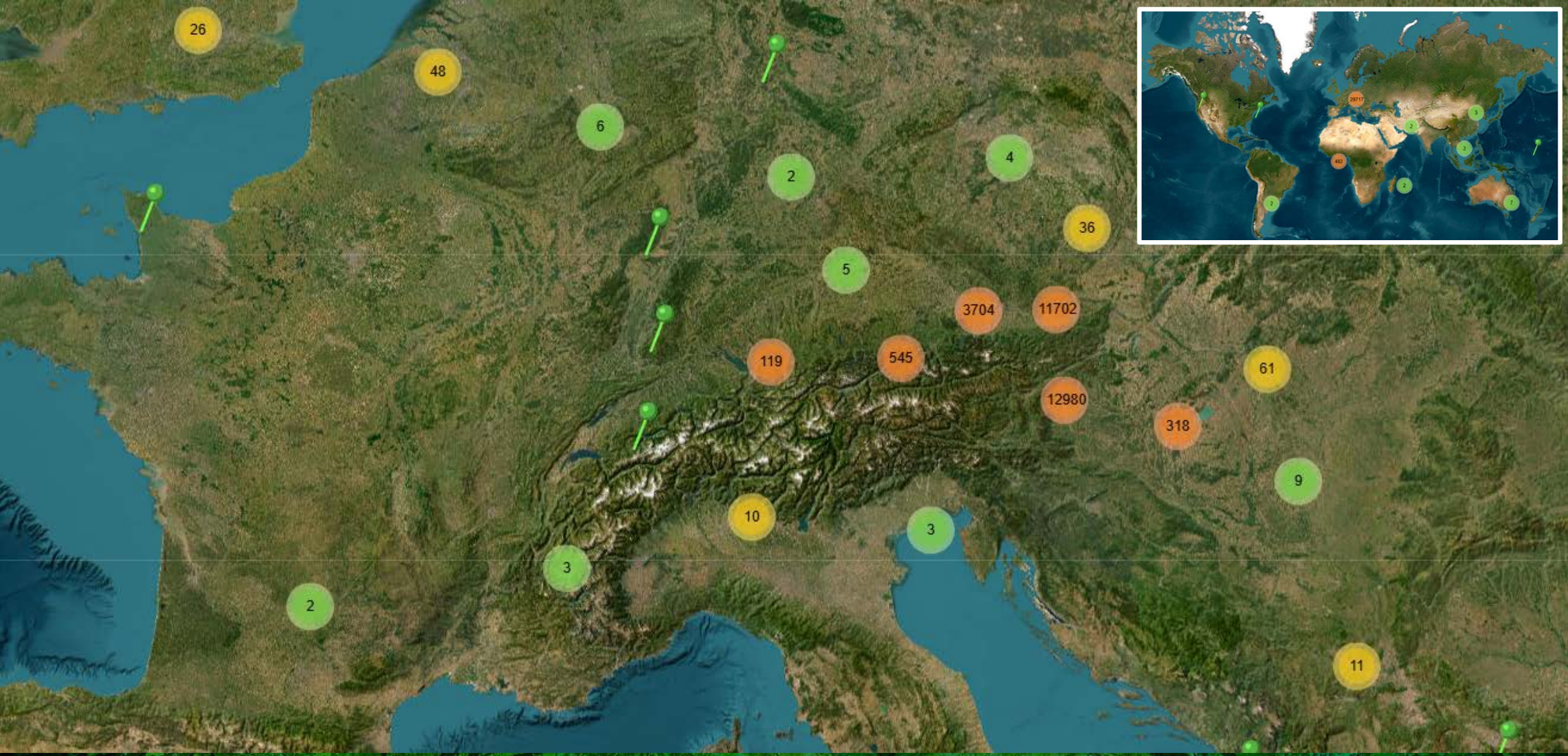


- the ACS method suggested by Walter Bitterlich (1947)
- based on the opening angle - Bitterlich stick
  - 1cm wide and 50cm long
  - counting trees wider than the stick width
- correction factor to convert to basal area
- AR helps visualize the stick width/opening angle









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Submissions
30762

Images
230920

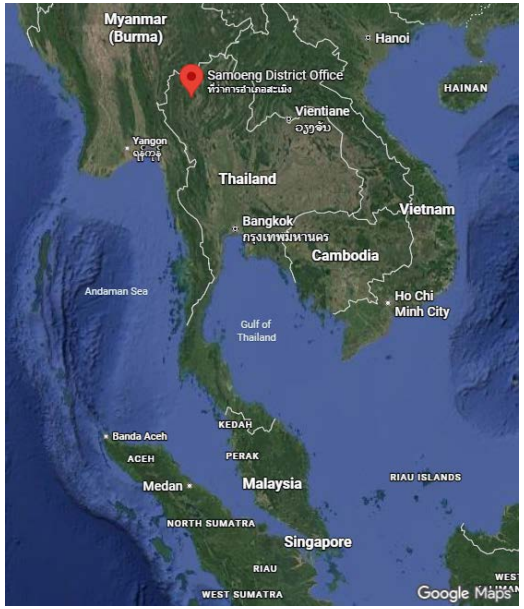
Users
428

Test Sites for Biomass Satellite Cal/Val

- Thailand, Chiang Mai Province
  - Forest of Samoeng District, part of Khun Khan National Park
  - Collaboration with Chiang Mai University
  - **Type of forest:** Mixed forest, dipterocarp forest, Hill evergreen forest, dry evergreen forest

- Kenya
  - Collaboration with Ferdinand Peters
  - Engagement with local community and university
  - Exploring fudging possibilities

- Any other areas?



Thank you!

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Error Propagation to Aboveground Biomass

Species	Equation	Source	DBH [cm]	TH [m]	P [g/cm³]	AGB Error [%]
Quercus robur	$\ln(AGB) = a + b \cdot \ln(DBH)$	(Zianis et al. 2005)	139.3	24.8	0.66	5.77
Quercus robur	$AGB = a \cdot (\rho \cdot TH \cdot \mathrm{DBH}^2)^b$	(Chave et al. 2014)	139.3	24.8	0.66	9.08
Fagus sylvatica	$\ln(AGB) = a + b \cdot \ln(DBH) + c \cdot \ln(TH)$	(Zianis et al. 2005)	54.2	12.7	0.56	15.52
Fagus sylvatica	$AGB = a \cdot (\rho \cdot TH \cdot \mathrm{DBH}^2)^b$	(Chave et al. 2014)	54.2	12.7	0.56	19.8
Pinus nigra	$AGB = a + b \cdot \mathrm{TH} \cdot \mathrm{DBH}^2 + c \cdot \mathrm{DBH}^2$	(Muukkonen and Mäkipää 2006)	75.45	20.4	0.41	12.48
Pinus nigra	$AGB = a \cdot (\rho \cdot TH \cdot \mathrm{DBH}^2)^b$	(Chave et al. 2014)	75.45	20.4	0.41	13.25